

Value of the Early Electroencephalogram After a First Unprovoked Seizure

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Key words

Electroencephalography, Predictive Value
Epilepsy, Adults
Seizure, First Unprovoked

ABSTRACT

Studies on the predictive value of the electroencephalogram (EEG) concerning the risk of seizure recurrence have shown contradictory results. We prospectively studied the predictive value of the standard EEG and EEG with sleep deprivation for seizure relapse in adult patients presenting with a first unprovoked seizure.

EEGs were performed on 157 adult patients within the first 48 hours of the first seizure. Additional EEGs with sleep deprivation were obtained in 60 cases. The standard EEG was abnormal in 70.7% and significantly associated with an increased risk of seizure recurrence [risk ratio 4.5, 95% confidence interval (CI) 1.8; 11.3, $p=0.001$]. Subgroup analysis revealed the highest recurrence rates for patients with focal epileptiform activity (risk ratio 2.2, CI 1.2; 4.2, $p=0.01$). EEGs with sleep deprivation were abnormal in 48.3% of all cases and revealed epileptiform discharges in 13.3% of the patients who had no epileptiform activity in the standard EEG.

Routine EEG revealed abnormalities in 60 of 94 patients who presented with normal neurologic status on admission. Further neuroradiological examinations detected previously unknown brain lesions in 19 of these cases, particularly cerebrovascular disease (CVD, $n=7$), brain tumors ($n=6$), and posttraumatic scars ($n=4$).

In conclusion, the EEG is important for the early detection of focal nonepileptic and epileptic abnormalities after a first unprovoked seizure in adult patients and may provide valuable information on previously unknown disorders, particularly CVD and cerebral tumors. The abnormal EEG is a highly significant predictor for seizure recurrence. An additional EEG with sleep deprivation is helpful in cases when standard EEG does not reveal epileptiform discharges.

INTRODUCTION

The predictive value of specific EEG abnormalities for the estimation of seizure recurrence in adult patients is not clear. Focal and generalized epileptiform activity, as well as focal nonepileptic discharges have been discussed.^{1,7}

Three studies, all in children, described a substantially increased recurrence risk for children with epileptiform EEG activity, whereas nonepileptiform abnormalities were insignificant.⁸⁻¹⁰ Further EEG studies found a higher risk of seizure recurrence associated with the presence of EEG abnormalities in adults and children.^{1-3,5,6,8-12} In these aforementioned studies, however, different methodological designs were used, such as retro- and prospective ascertainment. Most of the studies did not mention the interval between first seizure and initial EEG recording,^{1,2,4,7,11,13,14} and the classification of EEG abnormalities was not standardized^{1,4,13} or variable.^{2,3,5,7,11,14}

The aim of our prospective, hospital-based study was to evaluate the predictive value of standard EEG and EEG with sleep deprivation for seizure recurrence in adult patients after a first unprovoked seizure.

PATIENTS AND METHODS

Patients

Since 1991 we have been following a cohort of 157 adult patients, who were admitted to our department, in Mannheim, after a certain first unprovoked seizure. There was official permission for this study by the local authorities. An unprovoked seizure was defined according to the proposal of the International League against Epilepsy, encompassing remote symptomatic seizures as well as idiopathic and cryptogenic seizures.¹⁵ The definition of a single seizure includes status epilepticus and clusters of seizures, i.e., two or more seizures within 24 hours. Diagnosis was made based on the description of the episode and the results of the clinical examination, EEG and neuroimaging.

The patients were older than 16 years, and the first EEG was obtained within 48 hours of the first seizure (95% of the patients being seen within 24 hours). Patients were excluded from the primary analysis, if they (I) had a histo-

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Table 1

Categorization of standard EEG
and EEG with sleep-deprivation

0	Normal EEG
1	Nonspecific slowing (increased theta or delta activity adjusted for age)
2	Focal nonepileptic abnormalities (e.g., unilateral theta or delta focus)
3	Focal epileptiform activity (e.g., sharp waves, spikes) with or without secondary generalization
4	Generalized epileptiform activity (e.g., spikes and waves)

ry suspicious of previous epileptic seizures, (II) presented with absence or myoclonic seizures as initial seizure type, (III) if the first seizure was acute symptomatic, (IV) if they had a malignant brain tumor or (V) AIDS encephalopathy. Mean age was 48.1 ± 20.3 years (range 17 to 84 years); 97 of the patients were male (61.8%). The mean follow-up period was 34 months (range 3-88 months). Seven patients were lost to follow-up. All patients were untreated when their first EEG was performed. The goal of this study was to keep patients untreated after their first seizure. Treatment was always started after the seizure recurrence. A negligible number of patients however were started on medication by their practioners during further follow-up.

EEGs

All EEGs were recorded on 16-channel video-split-screen instruments with both referential and bipolar recordings by means of the International 10-20 electrode placement System. All EEGs were performed with hyperventilation for 3 minutes and intermittent photic stimulation. Sleep-deprived EEGs were performed on 60 patients whose initial EEG was normal or inconclusive, e.g., unspecific focal slowing. Sleep-deprivation EEGs were obtained after 3 to 7 days between 9:00 to 11:00 a.m. after a maximum of 3 hours of sleep the night before. EEGs were rated by two experienced neurophysiologists (the authors) according to a fixed protocol. The findings were classified in 5 categories (Table 1). If more than one abnormality was present, the highest value based on the categorization was used for analysis.

Statistics

The cumulative risks of recurrence were determined by life-table-methods with event defined as an unprovoked seizure recurrence.¹⁶ These methods take into account the length of follow-up in each individual. Univariate analysis (Kaplan-Meier survival curves) for each variable relates the cumulative percent recurrence to the time after first seizure.¹⁷ We used the proportional hazards model to estimate univariate rate ratios, defined as the ratio of the rate of seizure recurrence in the group of patients with a given factor to the rate of seizure recurrence without that factor.¹⁸

Findings were considered significant when the bounds of the 95% confidence interval did not include unity.

RESULTS

Seizure types: 95 patients (60.5%) had generalized tonic-clonic seizures without focal onset, of which 2 presented with status epilepticus. Nine (5.7%) had simple partial and 5 (3.2%) had complex-partial seizures. Secondly generalized seizures occurred in 27 patients (28.4%), 5 seizures (3.2%) remained unclassified.

Standard EEG: More than two-thirds of standard EEGs were abnormal (70.7%). Most frequent were focal abnormalities, i.e., a theta or delta focus (36.9%) or focal epileptiform activity (17.2%). We detected considerably low rates for generalized epileptiform activity (9.6%) and diffuse slowing (7.0%) (Table 2).

Forty-nine patients (31.2%) had a seizure recurrence (SR). Recurrence risks were estimated to be 23.1% after 6 months, 27.5% at one year and 31.0% at two years for all patients with a significantly increased risk, i.e., for patients with an abnormal EEG (Figure 1).

Of patients with SR, 10.2% had a normal standard EEG compared to 38.0% who remained seizure free. Focal epileptiform activity was detected significantly more frequent in patients with SR (26.5% vs. 13.0%). Focal nonepileptic abnormalities were associated more frequently, but not statistically significant, with an increased recurrence risk (46.9% vs. 32.4%). There were no significant differences in the rates of generalized epileptiform activity or diffuse slowing (Table 2).

In univariate analysis an abnormal EEG was significantly associated with an increased recurrence risk [risk ratio (RR) 4.5, 95% confidence interval (CI) 1.8; 11.3, $p=0.001$]. Subgroup analysis revealed the highest recurrence rates for patients with focal epileptiform activity (RR 2.2, CI 1.2; 4.2, $p=0.01$) and focal nonepileptiform abnormalities (RR 1.6, CI 0.9; 2.8, n.s.). Combined focal epileptiform and nonepileptiform activity were present in 54.1% (RR 2.8, CI 1.5; 5.3, $p=0.001$). Diffuse slowing and generalized epileptiform activity were not associated with significantly increased recurrence risks (Table 3).

Standard EEG was more sensitive for detecting abnormalities compared to both clinical and neuroradiological examinations on admission. The initial EEG was abnormal in 70.7%, the neurologic status in 40.1%, and CT-scan or MRI in 50.3%. Standard EEG revealed abnormalities in 60/94 patients who presented with normal neurologic status on admission. Most frequent were focal nonepileptiform abnormalities ($n=30$), focal ($n=14$) and generalized ($n=10$) epileptiform activity. Neuroradiological examination revealed previously unknown cerebral lesions in 19 of these cases, mainly brain tumors ($n=6$), ischemic stroke ($n=5$), trauma ($n=4$), and subcortical vascular encephalopathy ($n=2$). While 50% of the patients with cerebral neoplasm had a normal neurologic status, the EEG of

Table 2

EEG results among patients with and without seizure recurrence

	All patients no. (%)	Patients with SR no. (%)	Patients without SR no. (%)
Normal EEG	46/157 (29.3)	5/49 (10.2)	41/108 (38.0)
Unilateral focal abnormalities	58/157 (36.9)	23/49 (46.9)	35/108 (32.4)
Focal epileptiform activity	27/157 (17.2)	13/49 (26.5)	14/108 (13.0)
Gener. epileptiform activity	15/157 (9.6)	5/49 (10.2)	10/108 (9.3)
Diffuse slowing	11/157 (7.0)	3/49 (6.1)	8/108 (7.4)

SR = seizure recurrence

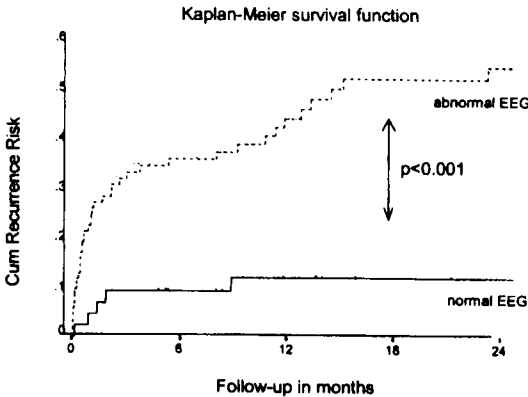


Figure 1.
Recurrence among patients with normal and abnormal EEG.

Table 3

Univariate risk ratios and 95% confidence intervals
[CI] estimated by Cox regression analysis

Variable	Risk ratio	95% CI	p
Abnormal EEG	4.5	1.8 - 11.3	$p = 0.001$
Focal slowing combined with focal epileptiform activity	2.8	1.5 - 5.3	$p = 0.001$
Focal epileptiform activity	2.2	1.2 - 4.2	$p = 0.01$
Focal nonepileptiform activity	1.6	0.9 - 2.8	n.s.
Generalized epileptiform activity	1.0	0.4 - 2.5	n.s.
Diffuse slowing	0.9	0.3 - 3.0	n.s.

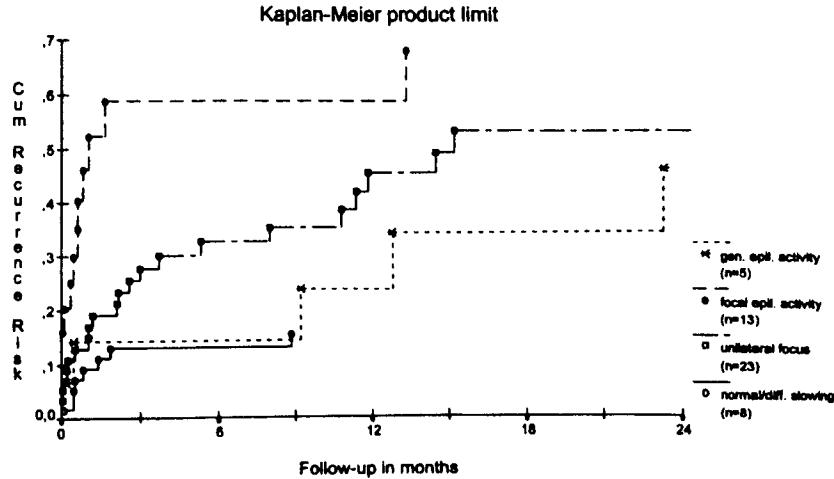


Figure 2.
Recurrence among
patients with different EEG
patterns

these patients was abnormal in 91.7%. In patients with cerebrovascular disease the EEG was abnormal in 88.6%, but only 31.4% had focal neurological deficits.

Sleep-deprivation EEG: EEG with sleep deprivation revealed focal spiking or sharp waves in 16.7% (compared to 17.2% in the standard EEG), and generalized epileptiform activity in 11.7% (9.6% of standard EEG). In 9 patients, EEG with sleep deprivation revealed abnormali-

ties previously not detected in routine EEG: epileptiform activity in 8 patients (5 focal, 3 generalized) and diffuse slowing in 1. Standard EEGs of these patients had been classified as normal in 3, diffusely slow in 2 and with focal nonepileptiform abnormalities in 4 cases. In 10 cases however, EEG with sleep deprivation did not reveal focal epileptiform and nonepileptiform abnormalities that had already been detected by the standard EEG (Table 4).

Table 4

Number of abnormalities detected in standard EEG and EEG with sleep deprivation		
	Standard EEG [Number of abnormalities not detected in EEG with sleep-deprivation]	Sleep-deprivation EEG [Number of abnormalities not detected in standard EEG]
Unilateral focal abnormalities	5	0
Focal epileptiform activity	5	5
Gener. epileptiform activity	0	3
Diffuse slowing	3	1

DISCUSSION

The predictive value of the EEG after a first unprovoked seizure has proved to be a controversial topic. The controversies are based on different aspects: study design (e.g., prospective vs. retrospective, new-onset epilepsy vs. first-seizure method), age, site of recruitment, or time point of study entry. This led to considerable variations in estimation of recurrence rates and identification of predictors for seizure relapse.^{1,3,6,11,13,19} There have been only a few studies based on the assessment at the time of their first seizure.^{2,8} The cumulative risk of seizure recurrence in our study is consistent with the results of previous authors^{1,2,8,11,20} and a recent meta-analysis²¹ for studies using a prospective ascertainment and a first-seizure method.

The rate of EEG abnormalities was 70% in our study. This rate corresponds well with previous results that range from 41 to 80% overall and from 9 to 63% in selected populations, e.g., patients with idiopathic or remote symptomatic seizures.^{1,3,8-12,22} An abnormal EEG was predictive for a significantly increased recurrence risk in all^{1,3,8-10,12} but two^{5,11} studies. However, the interval between the first seizure and EEG recording, which may be decisive for sensitivity, was not mentioned in previous studies (in two studies the interval was > 48 hours and < 10 days).^{3,5} This has recently been criticized. In our study standard EEG was performed within 48 hours of the first seizure. In our opinion, this short interval explains the substantial rate of abnormalities and the increased recurrence risk associated with an abnormal EEG, a risk considerably higher than in previous studies.^{1,3,8-12} A loss of epileptic excitability with increasing interval from the first seizure might be the reason for the lower rate of abnormalities in the sleep deprivation EEG in our study (48.3% vs. 70.7%). A similar change was also reported by van Donselaar et al.,⁷ especially regarding focal epileptic and nonepileptic discharges. However, EEG with sleep deprivation may provide useful information in patients without epileptiform

activity in standard EEG,⁷ and in these cases it should be performed as soon as possible.

Data on specific EEG abnormalities associated with higher recurrence rates are rare. The high predictive value for SR of primary generalized spike-waves was reported only in a selected population of patients who most likely suffered from idiopathic epilepsy.^{2,7} Our results are comparable with previous studies, which reported the highest risks of recurrence for patients with focal epileptiform discharges and focal nonepileptic EEG abnormalities.⁷⁻¹⁰ The low rate of generalized epileptiform activity and idiopathic seizures in our study can be explained by the inclusion criteria: patients with petit mal seizures or a history suspicious of epileptic attacks, which frequently occur in idiopathic epilepsies, were not enrolled.

High excitability, the substantial rate of epileptiform activity and the significant association with recurrent seizures emphasizes the necessity of consulting a specialist and the need for an EEG examination immediately after a first unprovoked seizure. We conclude that a standard EEG, if performed within 48 hours, is crucial for the detection of focal nonepileptic and epileptic abnormalities after a first unprovoked seizure. In a substantial number it may help to detect previously unknown disorders, particularly cerebrovascular disease and cerebral tumors. Although the EEG may be an indicator of underlying brain lesions, it will not become a substitute for imaging studies. Both functional and structural tests are appropriate in these patients and provide valuable complementary information.

CONCLUDING REMARKS

An early EEG performed within 48 hours of the first seizure is a highly significant predictor for seizure recurrence. Additional EEGs with sleep deprivation may be helpful whenever standard EEG does not reveal epileptiform discharges. As a consequence to our results, we will prospectively include 24 hour continuous EEG-monitoring to obtain even more information on disturbed brain function immediately after a first seizure.

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